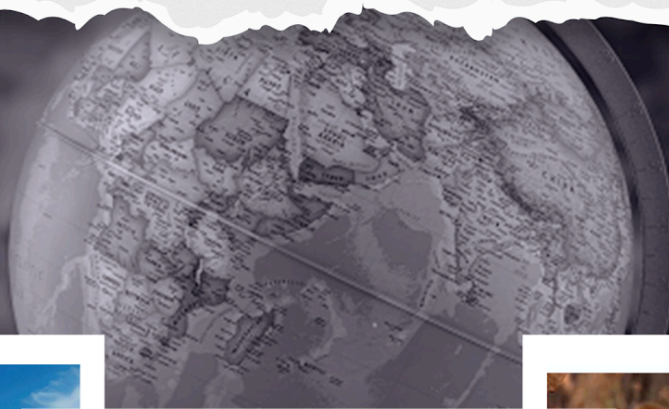


General Knowledge **World Geography**

STUDY NOTES

Motions of the Earth



Motions of the Earth

Introduction

The Earth undergoes two fundamental types of motion:

- Rotation
- Revolution

These motions are critical in shaping the **planet's climatic conditions, length of days and nights, and seasonal variations.**

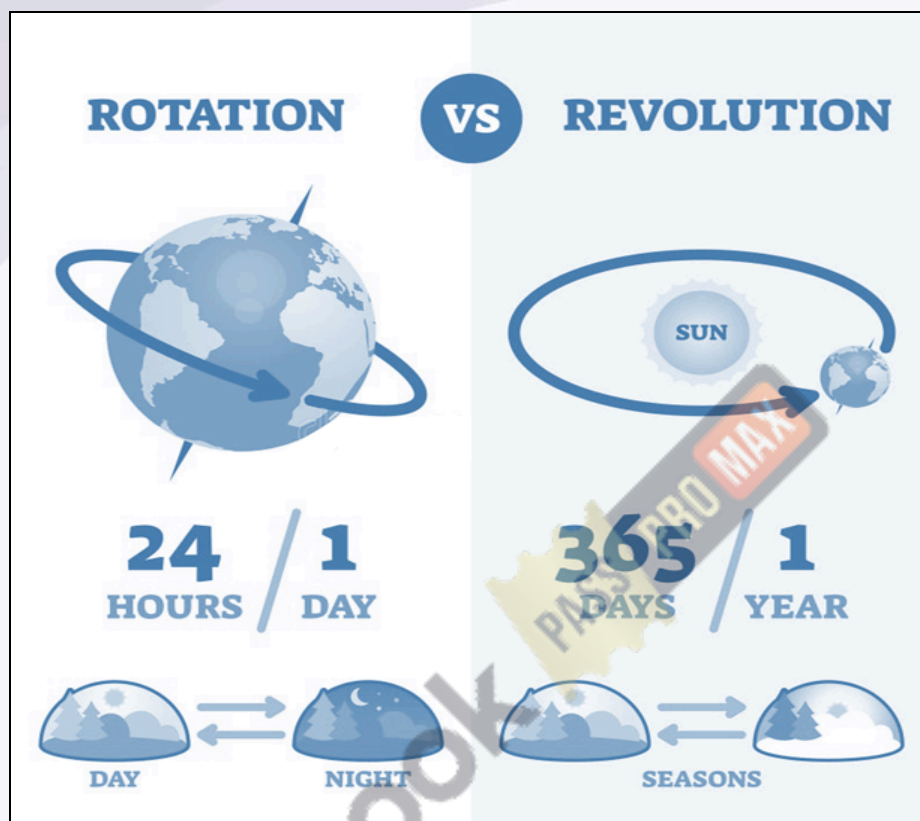


Rotation

- The Earth rotates on an imaginary axis that runs through its poles.
- The Earth's axis is tilted at **$66\frac{1}{2}^{\circ}$ with respect to the plane of its orbit (orbital plane).**
- One rotation takes **approximately 24 hours**, termed an Earthday.
- **Effects:** Causes day and night.

Revolution

- The Earth revolves around the Sun in an elliptical orbit.
- The revolution takes $365\frac{1}{4}$ days (1 year).



Leap Year:

- To account for the **extra 6 hours annually**, one extra day is added every four years, resulting in a year with **366 days (Leap Year)**.
- Example: February has 29 days in a leap year.

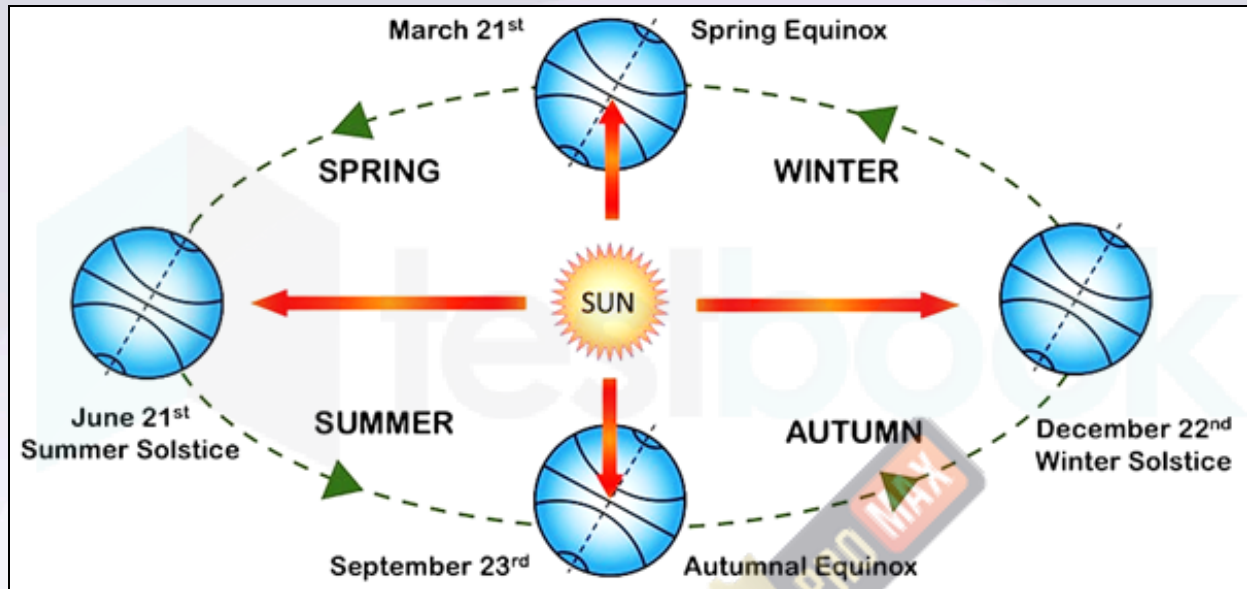
Circle of Illumination

- The dividing line between the **daylit and dark halves** of the Earth.
- Does not align with the Earth's axis due to its tilt.

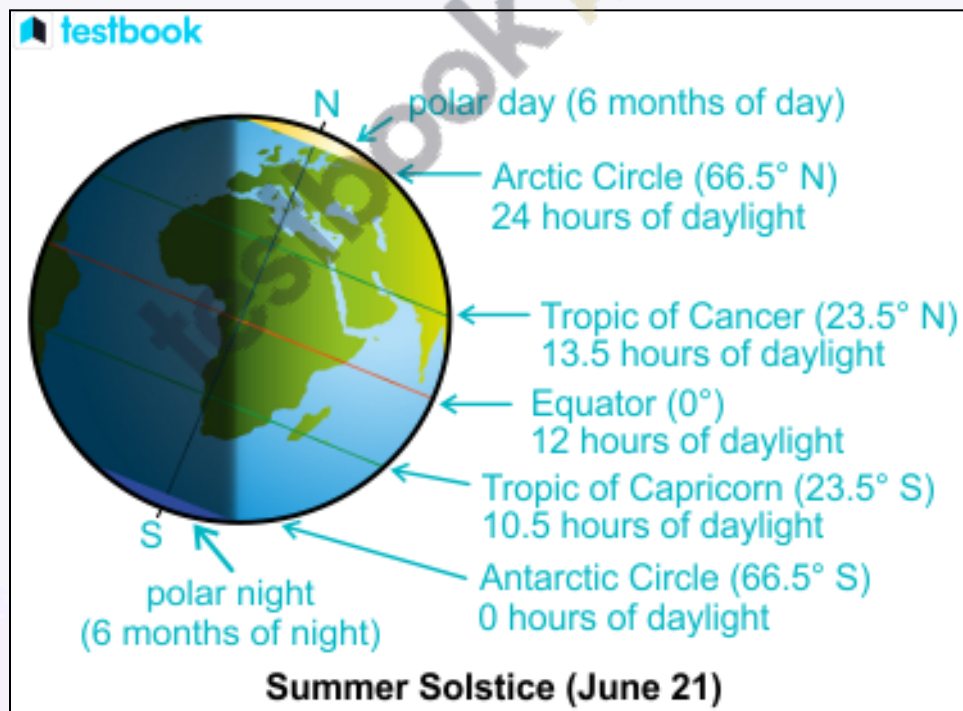
Key Phenomena Related to Earth's Motions

- The Earth's tilt and revolution cause different seasons
- A year is usually divided into **summer, winter, spring, and autumn seasons**.

- Seasons change due to the change in the **position of the earth around the sun**.



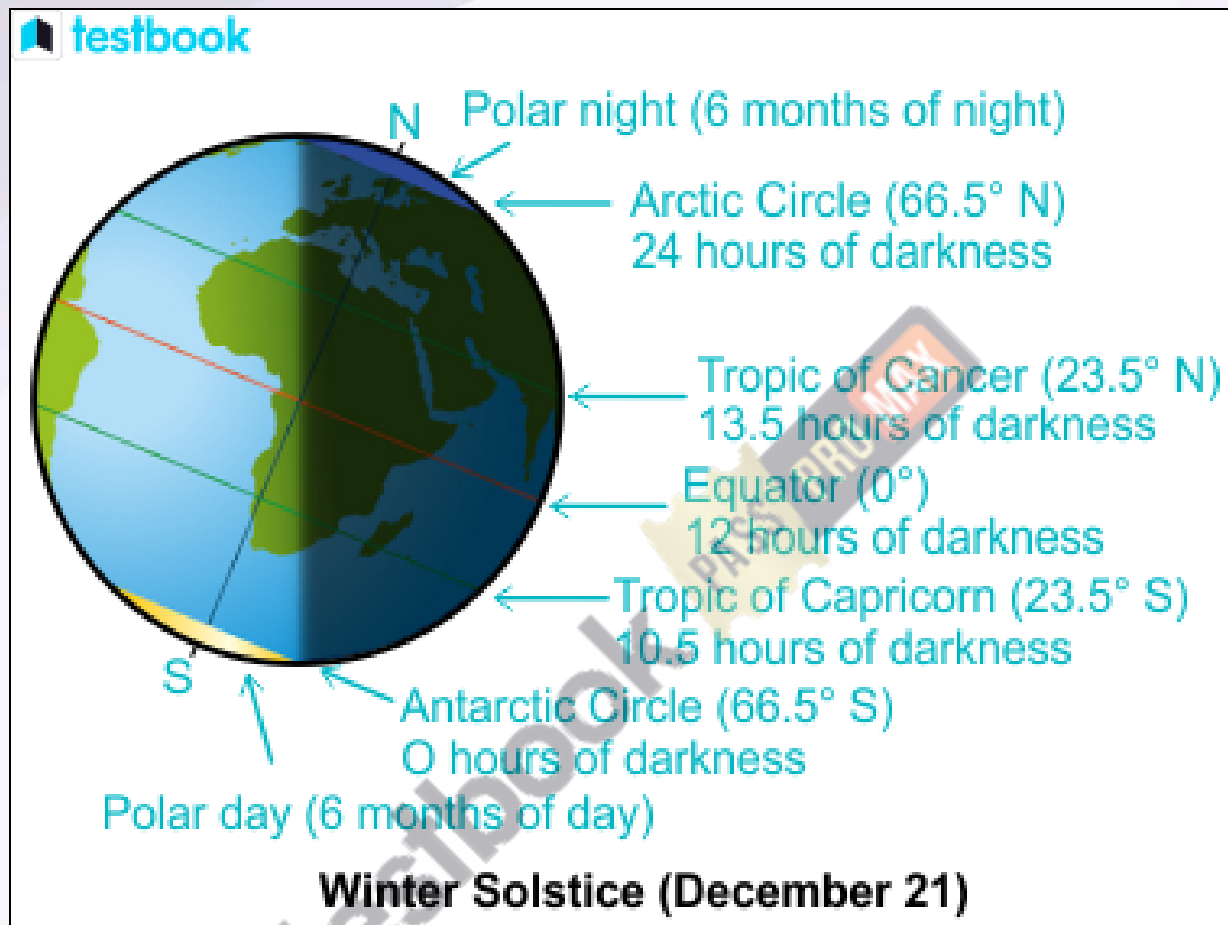
Summer Solstice (21st June):



- The Northern Hemisphere tilts towards the Sun.
- The **Tropic of Cancer** receives **direct sunlight**.
- Longest day and shortest night** in the Northern Hemisphere.

- Opposite conditions (winter) occur in the Southern Hemisphere.

Winter Solstice (22nd December):



- The Southern Hemisphere tilts towards the Sun.
- The Tropic of Capricorn receives direct sunlight.
- Longer days and shorter nights in the Southern Hemisphere (summer).
- Opposite conditions occur in the Northern Hemisphere.

Equinox (21st March and 23rd September):

- Sunlight falls directly on the equator.
- Equal length of day and night globally.
- Marks the onset of **spring or autumn** depending on the hemisphere.

Difference between Equinox and Solstice

Equinox	Solstice
The period of the year when the Sun is nearest to the equatorial region, resulting in equal durations of day and night.	The time during the year when the Sun is furthest away from the equator, resulting in prolonged nights and days.
It takes place at the beginning of the spring and fall seasons.	It occurs two times a year, once in summer and once in winter.
The two equinoxes occur on March 21st and September 23rd, respectively.	The two solstices occur on June 21st or 22nd and December 21st or 22nd, respectively.
The equator is located near the subsolar point.	On June 22nd, the subsolar point is 23.5° North, while on December 22nd, it is 23.5° South.
The vernal equinox, named after the Latin words <i>aequi</i> (equal) and <i>noctium</i> (night), marks the start of spring.	The term is derived from the Latin words <i>sol</i> (sun) and <i>stitium</i> (standing still), signifying the transition between day and night.
The Sun forms a 90-degree angle with the equator.	The Sun forms a 23.5-degree angle with the Earth's equator.

Day and Night at the Poles

- Due to the axial tilt, the **poles experience six months of daylight followed by six months of darkness.**
- On 21st June, the **Arctic Circle experiences 24 hours of daylight** as the North Pole is fully exposed to the Sun.
- On 22nd December, **the Antarctic Circle experiences 24 hours of daylight** due to the South Pole's exposure to the Sun.